

Weighted first derivatives of the full pair inverse

HOCF–RIESZ, Round 007

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Scope and status. This is a synthesis of the candidate THM–027. Its complete proof and separate independent audit dispositions are recorded in the immutable Markdown reports and canonical theorem ledger. The construction establishes a fixed- N first-derivative prerequisite; it does not establish an interacting-particle Itô drift or uniform fluctuation estimates.

Precise assertion

Use the unit Haar torus, coefficient-one periodic Riesz potential g , and $K = -\nabla g$. Fix $d \geq 3$, $0 < s \leq d - 2$, $N \geq 2$, $T < \infty$ and $0 \leq \nu \leq \nu_* < \infty$. Prescribe time-continuous data $\mu_t \in C^2$, $u_t \in C^1$ and $f_t \in C^3$, with uniform norms; μ_t is a probability density. General unknown mean-field norms are not asserted. The actual homogeneous Fourier reference and smooth terminal test from Round005 meet these hypotheses.

The pair generator, source and full inverse are

$$\begin{aligned} G_t &= \nu(\Delta_x + \Delta_y) + u_t(x) \cdot \nabla_x + u_t(y) \cdot \nabla_y + \frac{1}{N} K(x - y) \cdot (\nabla_x - \nabla_y), \\ J_t &= K(x - y) \cdot (\nabla f_t(x) - \nabla f_t(y)), \\ \Phi_t &= U_t + \int_t^T S_{t,a} R_a \Phi_a da, \quad U_t = \mathbb{E}_{t,x,y} \int_t^T J_a(Z_a) da, \quad R = R_x + R_y. \end{aligned}$$

Here S is the already constructed noncolliding auxiliary pair evolution and Φ the unique bounded Borel terminal-zero full inverse. Its previously proved Haar L^2 estimate alone supplied no derivative.

Fix $1 < q < d/2$. Choose a positive smooth off-diagonal weight $w_q(z)$ equal to $|z|^{-q}$ on an embedded ball and bounded above and below outside that ball. The new assertion is

$$\sup_{0 \leq t \leq T} \left(\|\Phi_t\|_\infty + \sup_{x \neq y} \frac{|\nabla_{x,y} \Phi_t(x, y)|}{w_q(x - y)} \right) \leq C < \infty.$$

The representative is C^1 off the diagonal, its derivatives are jointly Borel, and they are global weak derivatives. Thus $\sup_t \|\Phi_t\|_{H^1} < \infty$. The constant may depend on N, ν_*, T, d, s, q and the displayed data and cutoffs. It is uniform in the chosen ν in that interval and may grow rapidly in N .

One-sided growth and weighted occupations

Locally put $z = x - y$ and $r = |z|$. Then $K(z) = s|z|^{-s-2}z + k(z)$ with smooth odd k . Put $k_0 = 2s/N$. The nonsingular relative drift is bounded by $L_0 r$ for a finite fixed-data constant L_0 . The principal pair Jacobian has zero center eigenvalues and difference-space eigenvalues

$$k_0 r^{-s-2} \quad \text{transverse}, \quad -(s+1)k_0 r^{-s-2} \quad \text{radial}.$$

Its largest eigenvalue is the transverse one. Replacing it by the absolute matrix norm loses the required range. Choose a nonnegative global function ℓ equal to $k_0 r^{-s-2} + L$ near zero and dominating the largest symmetric drift-Jacobian eigenvalue elsewhere. The flow derivative $A_{t,a}$ then obeys

$$\|A_{t,a}\| \leq \exp \left(\int_t^a \ell(Z_r) dr \right)$$

along each noncolliding path. This pathwise fact alone does not justify expectation differentiation.

Choose weights with the exact product property $w_\alpha^p = w_{p\alpha}$: take $R = 1/16$ and a smooth radial cutoff χ equal to one on B_R and supported in B_{2R} , and set $w_\alpha = \exp(\alpha\chi \log(1/r))$, extended as one outside the chart. For $p \geq 0$ and $\alpha > p$, the inner-ball calculation is

$$\frac{(G_t + p\ell)w_\alpha}{w_\alpha} \leq -(\alpha - p)k_0 r^{-s-2} + 2\nu_* \alpha(\alpha + 2 - d)_+ r^{-2} + \alpha L_0 + pL.$$

Write $b = (\alpha - p)k_0 > 0$ and $a = 2\nu_* \alpha(\alpha + 2 - d)_+$. The positive diffusion contribution is controlled by the explicit maximum

$$\sup_{v \geq 0} \left(av^2 - \frac{b}{2} v^{s+2} \right) = \frac{s}{s+2} a \left(\frac{4a}{(s+2)b} \right)^{2/s},$$

with value zero when $a = 0$. The more singular negative repulsive term is essential, including in dimension three where the displayed diffusion term can be positive. Bounded annular cutoff derivatives give a finite specified $C_{p,\alpha}$ such that

$$(G_t + p\ell)w_\alpha \leq C_{p,\alpha} w_\alpha - \frac{b}{2} \mathbf{1}_{r \leq R} r^{-\alpha-s-2}.$$

The full Markdown proof lists every annular and data constant. They can already grow like $N^{2/s}$ inside a later time exponential.

Stopped smooth Itô applied to the nonnegative weighted process gives, after Fatou for its terminal term and monotone convergence for occupation,

$$\begin{aligned} \mathbb{E} e^{p \int_t^{t+h} \ell} w_\alpha(Z_{t+h}) &\leq e^{C_{p,\alpha} h} w_\alpha(z), \\ \mathbb{E} \int_t^{t+h} e^{p \int_t^a \ell} \mathbf{1}_{r_a \leq R} r_a^{-\alpha-s-2} da &\leq \frac{2}{b} e^{C_{p,\alpha} h} w_\alpha(z). \end{aligned}$$

No uniform integrability of stopped weights or zero collision boundary value is assumed. Apply the level bound for a nonnegative supermartingale at doubled parameters. Integrating its square-root tail gives

$$\mathbb{E} \sup_{t \leq a \leq t+h} e^{p \int_t^a \ell} w_\alpha(Z_a) \leq 2e^{C_{2p,2\alpha} h/2} w_\alpha(z).$$

The expectation derivative is a separate proof step

For a fixed driving path let F_m be the minimum of m and the maximal weight w_1 along its local trajectory through the horizon, using infinity at a finite collision lifetime. These truncated functions are locally Lipschitz in the starting point. Below the truncation level this follows from one safe smooth cutoff flow. Above it a finite time has already crossed the level, so F_m is locally constant, even at an exploding start.

The previous maximal estimate bounds F_m and its local weak gradient in L^P of the starting point and probability, uniformly in m , for every $P > 2d$. The gradient is bounded by a constant times $\sup_a e^{\int_t^a \ell} w_2(Z_a)$. Averaging the line-segment fundamental theorem over a coordinate ball gives the elementary supremum estimate

$$\|v\|_{L^\infty(B_b)} \leq C_{b,P,d} (\|v\|_{L^P(B_{2b})} + \|\nabla v\|_{L^P(B_{2b})}).$$

Its kernel is $|x - y|^{1-2d}$, whose conjugate-power integral is finite because $P > 2d$. Fubini and Fatou bound a subsequence of these random norms almost surely. Monotonicity of F_m then bounds the untruncated path weight on the smaller initial ball. No start in that ball can explode. A countable cover yields a simultaneous local flow for each fixed initial time and horizon; its image of a compact initial ball stays uniformly separated from collisions. This common-flow conclusion is derived, not imported.

For a pathwise C^1 functional $H(z, W)$, a uniform local bound $\sup_z \mathbb{E}|D_z H|^p < \infty$ with $p > 1$ bounds finite differences by the line-segment formula and Jensen. Their tails are at most a constant times K^{1-p} , giving uniform integrability. This proves $DEH = EDH$ and continuity of the derivative. The weights above provide exactly those bounds for propagated terminal functions.

For the singular source, direct differentiation gives $|\nabla J| \leq C_J r^{-s-1}$ near zero, with bounded remainder. Choose $\alpha > p$ so that $\alpha + s + 2 \geq p(s + 1)$. The retained negative occupation term bounds the p th moment

of $\int_t^T \|A_{t,a}\| |\nabla J_a(Z_a)| da$. For example $p = 2$, $\alpha = 2(s+2) + 1$ suffice for uniform integrability. Thus the actual signed source potential satisfies

$$\nabla U_t = \mathbb{E} \int_t^T A_{t,a}^\top \nabla J_a(Z_a) da, \quad \sup_t \sup_E \frac{|\nabla U_t|}{w_q} \leq C_J e^{C_{1,q}T} \left(T + \frac{2}{(q-1)k_0} \right).$$

The zero-terminal derivative is included. The previously proved bounded source-potential estimate supplies its separate zeroth-order norm.

Both responses and the full inverse

The exact first-slot response is

$$R_x F = - \int \mu(x+w) F(x+w, y) D(dw) - \int K(w) \cdot \nabla \mu(x+w) F(x+w, y) dw, \quad D = \operatorname{div} K.$$

For $0 < a, q < d$, splitting near the two separate singular points and their complement proves

$$\int \chi(w) |w|^{-a} w_q(w+z) dw \leq C_{a,q} w_q(z), \quad z \neq 0.$$

The convolution need not remain bounded when $a+q \geq d$; the weighted bound uses $a < d$ and $q < d$ separately. It applies to the force and, below Coulomb, the divergence density. At Coulomb retain exactly $D = c_d(\delta_0 - dw)$; the atom samples the present off-diagonal point.

A bounded C^1 function with gradient bounded by w_q has global weak first derivatives: deletion of a diagonal tube has boundary contribution $O(\|F\|_\infty \delta^{d-1})$, while its gradient is integrable. Differentiate the translated products in the response formula in this weak sense. Fubini, the convolution bound and the C^2 density norms control every term. Continuity off the diagonal follows by separating the two singular points; near the moving second point a change of variable fixes it before dominated convergence. The second response is treated in its own slot. Thus for $\|F\|_{\mathcal{X}_q} = \|F\|_\infty + \sup |\nabla F|/w_q$,

$$\|R_t F\|_{\mathcal{X}_q} \leq C_{R,q} \|F\|_{\mathcal{X}_q}, \quad \|S_{t,a} F\|_{\mathcal{X}_q} \leq e^{C_{1,q}(a-t)} \|F\|_{\mathcal{X}_q}.$$

Every density-gradient and second-density-derivative term is retained.

Write $(VF)_t = \int_t^T S_{t,a} R_a F_a da$ and $B_q^* = \sup_t \|U_t\|_{\mathcal{X}_q} < \infty$, as proved by the source estimates. The ordered Volterra series then has bounds

$$\|(V^k U)_t\|_{\mathcal{X}_q} \leq B_q^* e^{C_{1,q}(T-t)} \frac{[C_{R,q}(T-t)]^k}{k!}.$$

It converges uniformly in values and locally uniformly in derivatives, without a separable weighted-sup-space assumption. Bounded Borel uniqueness identifies its sum with the genuine full inverse. Finally $2q < d$ gives

$$\sup_t \|\Phi_t\|_{H^1}^2 \leq C^2 \left(1 + \int w_{2q} \right) < \infty.$$

No diagonal trace follows. The crude internal-drift majorant is r^{-s-1-q} , which is not integrable at Coulomb for $q > 1$. Second/time derivatives or a directional cancellation, the exact interacting particle Itô identity and uniform evolved residual/bracket estimates remain separate open gates.